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Geometric Dark Energy: Phenomenological Viability, Systematic Closures, and Requirements for Completion

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We present a phenomenological dark-energy framework rooted in Einstein-Cartan-Holst (ECH) gravity, report the systematic closure of four minimal and one near-minimal first-principles route to deriving the late-time vacuum term, and extract the structural requirements that any viable geometric dark-energy completion must satisfy.

The minimal ECH framework—with algebraic torsion, fixed Barbero-Immirzi parameter $\gamma = 0.274$, and minimal Dirac coupling—provides a consistent phenomenological ansatz that reduces fine-tuning from 10^{120} to $\sim 10^{5}$ and yields MCMC fits partially reducing the H_0 and σ_8/S_8 tensions. However, four systematic attempts to derive the late-time $w = -1$ behavior within the minimal model have all failed, sharing a common structural origin: algebraic torsion washes out after elimination, leaving no infrared fingerprint. Additionally, the most natural next-generation extension—propagating torsion in Poincaré gauge theory (PGT)—reveals a new obstruction: in the ghost-free 0^- axial mode, the mass and matter coupling are locked ($m \propto g_{\rm eff} \propto 1/\sqrt{|t_3|}$), so making the mode cosmologically light simultaneously drives its coupling $\sim 10^{29}$ times below gravitational strength, rendering it observationally inert. No symmetry within PGT protects this mass.

From these closures we extract six structural lessons and formulate requirements that constrain the space of viable geometric dark-energy models: the theory must survive reduction, address scale naturalness, offer a distinctive observable, and—crucially—avoid the mass-coupling lock in which a light-mass limit automatically decouples the field from matter.

The framework's phenomenological viability (cosmological fits, scaling argument, fine-tuning reduction) survives all closures intact. Its scientific contribution is the combination of a well-tested phenomenological model with a systematic map of the theory landscape showing what does not work and why, narrowing the path for any future geometric completion.

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I. INTRODUCTION

The cosmological constant problem—why the observed vacuum energy density $\rho_\Lambda \sim (2.3\ \mathrm{meV})^4$ is $\sim 10^{120}$ times smaller than the naive quantum field theory estimate $\sim M_{\mathrm{Pl}}^4$ [1]—remains one of the deepest unsolved problems in fundamental physics. Growing tensions within ΛCDM, including the $\sim$4.9σ Hubble tension [2, 3] and 2–3σ σ_8/S_8 tension [4, 5], together with hints of dynamical dark energy from DESI DR2 BAO measurements [6, 7], motivate exploration of geometric approaches to dark energy.

Einstein-Cartan-Holst (ECH) gravity—Einstein-Cartan theory supplemented by the Holst term with Barbero-Immirzi parameter γ—is a natural starting point for geometric dark energy. The theory is well-motivated by loop quantum gravity [8], introduces spin-torsion coupling with parity-violating structure [9, 10], and provides a non-singular quantum bounce [12]. The question is whether this theoretical structure can generate a natural origin for late-time dark energy.

This paper reports three results. First, we present a phenomenological dark-energy framework built on ECH gravity that is observationally consistent and reduces fine-tuning significantly, with MCMC fits partially reducing the H_0 and σ_8 tensions. Second, we report the systematic failure of four minimal first-principles routes to derive the late-time $w = -1$ behavior. Third, we report an assessment of the most natural next-generation extension—propagating torsion in Poincaré gauge theory (PGT)—finding that it is structurally viable but phenomenologically empty for dark energy due to a *mass-coupling lock*: the same parameter that makes the torsion mode cosmologically light also suppresses its matter coupling to $\sim 10^{29}$ times below gravitational strength, rendering it observationally inert. From these results we extract six structural lessons and formulate concrete requirements for any viable geometric dark-energy completion.

The combination establishes the current scientific status of the program:

- The minimal ECH framework is a *motivated phenomenological model*, not a first-principles derivation of dark energy.
- The minimal model and its most natural extension have both been tested and found insufficient, for distinct structural reasons.
- The closures map the theory landscape and identify nontrivial requirements—including avoidance of the mass-coupling lock—that constrain future geometric dark-energy proposals.

A. Relationship to prior work

This paper supersedes an earlier version [20] that presented the same phenomenological framework but was organized around the minimal ECH model as if it were a sufficient foundation. The present version incorporates the results of a structured derivation program (documented in a companion technical note [21]) that tested four minimal routes and found all to be negative. The phenomenological results are unchanged; the framing, interpretation, and forward-looking directions are substantially revised.

II. EINSTEIN-CARTAN-HOLST FRAMEWORK

A. Action and field content

The starting point is the Einstein-Cartan-Holst action minimally coupled to N_f Dirac fermions:

$$S_{\mathrm{ECH}} = \frac{M_{\mathrm{Pl}}^2}{2}\int \left[\varepsilon_{IJKL} + \tfrac{2}{\gamma}\,\eta_{I[K}\eta_{L]J}\right] e^I \wedge e^J \wedge F^{KL}[\omega] + \int d^4x\,|e|\left[\bar\psi\, i\gamma^\mu D_\mu\psi - m\bar\psi\psi\right], \tag{1}$$

with $D_\mu = \partial_\mu + \frac{1}{4}\omega_\mu^{IJ}\gamma_I\gamma_J$ the Lorentz-covariant derivative and $\gamma = 0.274 \pm 0.020$ the Barbero-Immirzi parameter fixed by loop quantum gravity [8].

The first term in Eq. (1) is the Holst action: the Palatini (first-order) formulation of general relativity supplemented by the Holst term $(1/\gamma)\varepsilon_{IJKL}e^I e^J F^{KL}$, which is parity-odd and classically equivalent to a topological term (the Nieh-Yan density) on shell. The second term is the standard minimal Dirac coupling.

B. Torsion equation of motion

In ECH gravity, the spin connection ω_μ^{IJ} is an independent dynamical variable, and its equation of motion is algebraic—torsion is *not propagating*. Solving the torsion equation exactly decomposes the connection as $\omega = \mathring{\omega} + K$, where $\mathring{\omega}$ is the torsion-free Levi-Civita connection and K is the contorsion tensor determined algebraically by the fermion spin density [9, 14].

C. Reduced action after torsion elimination

Substituting the algebraic torsion solution back into the action yields the reduced action [9, 13]:

$$S_{\rm red} = S_{\rm EH}[g] + S_{\rm Dirac}[g,\psi] + S_{\rm 4f}[\psi;\gamma], \qquad (2)$$

where $S_{\rm EH}$ is the standard Einstein-Hilbert action, $S_{\rm Dirac}$ is the standard minimally-coupled Dirac action with the Levi-Civita connection, and $S_{\rm 4f}$ is the torsion-induced four-fermion contact interaction:

$$\mathcal{L}_{\rm 4f} = -G_{\rm eff}\left(\bar{\psi}\gamma^\mu\gamma^5\psi + \tfrac{1}{\gamma}\,\bar{\psi}\gamma^\mu\psi\right)^2, \quad G_{\rm eff} = \frac{3\kappa^2}{16}\,\frac{\gamma^2}{\gamma^2+1}. \qquad (3)$$

This perfect-square structure [21] constrains the Fierz channel decomposition. At $\gamma = 0.274$, the vector admixture $1/\gamma \approx 3.65$ dominates; the scalar/pseudoscalar channel coupling $G_{\rm SP} \propto (\gamma^2-1)/(\gamma^2+1)$ is repulsive (negative).

a. Critical structural observation. After torsion elimination, *all* γ-dependence resides in the four-fermion term $S_{\rm 4f}$. The Dirac kinetic operator $i\gamma^\mu\mathring{\nabla}_\mu$ is the standard Levi-Civita operator, independent of γ. This structural fact is the root cause of the failures reported in Sec. V: the Holst sector's content does not survive torsion elimination in any form that can generate late-time vacuum physics at one loop or below.

III. PHENOMENOLOGICAL DARK-ENERGY ANSATZ

A. The scaling argument

The ECH framework contains a parity-odd vacuum operator $\varepsilon^{abcd}K_{ab}R_{cd}$ sourced by the fermion spin density via torsion. At early times (Planck-density bounce), this operator has a natural magnitude set by $\alpha/M \sim \mathcal{O}(\kappa^2)$, where α/M is the parity-odd coefficient in the effective Lagrangian.

Inflationary expansion after the bounce dilutes this operator by a factor $\mathcal{D}_{\rm inf} \sim e^{-3N_{\rm tot}}$, where $N_{\rm tot} \sim 92$ e-folds. The resulting late-time dark-energy density is parametrically

$$\rho_\Lambda \approx \Xi\, M_{\rm Pl}^4, \qquad \Xi \equiv \left[(\alpha/M)\, M_{\rm Pl}\right]\mathcal{D}_{\rm inf}, \qquad (4)$$

giving $\Xi \sim 10^{-122}$ for appropriate values of α/M and $N_{\rm tot}$.

a. What this is. Equation (4) is a *phenomenological scaling ansatz.* It traces a dimensional chain from the Planck scale through inflationary dilution to the observed vacuum energy. It reduces fine-tuning from 10^{120} (the cosmological constant problem in ΛCDM) to $\sim 10^5$ (sensitivity to $N_{\rm tot}$ within $\Delta N \approx 4$ e-folds).

b. What this is not. Equation (4) is *not* a derivation of $w = -1$. The parity-odd operator is sourced by the spin density, which vanishes at late times ($K_{ab} \to 0$). For the residual to persist as a true vacuum energy, one must show that integrating out the spin degrees of freedom generates an IR-constant term in the effective action. Four attempts to establish this have failed (Sec. V). We therefore treat $w = -1$ as a phenomenological assumption, consistent with observations ($|w+1| \lesssim 10^{-2}$ from DESI+Planck).

B. Cosmological parameters

The phenomenological framework motivates a ΛCDM+$\Delta N_{\rm eff}$ extension: the bounce scenario qualitatively predicts additional relativistic species from particle production near the Planck density, but quantitative estimates are not yet available. The extension adds one effective parameter ($\Delta N_{\rm eff}$) to ΛCDM; a second parameter ($(\omega/H)_0$) is bounded by CMB isotropy to $< 5\times 10^{-11}$ [24] and contributes negligibly.

1. Datasets

We analyze four dataset combinations of increasing constraining power:

1. *Planck-only*: Planck 2018 NPIPE, specifically `plikHM_TTTEEE` + `lowl` + `lowE` + `lensing` [2].
2. *Planck + BAO*: Adding DESI 2024 DR1 (seven redshift bins, $z_{\rm eff} = 0.30$–2.33) [6], supplemented by 6dFGS ($z_{\rm eff} = 0.106$) and SDSS MGS ($z_{\rm eff} = 0.15$).
3. *Planck + BAO + SN*: Adding Pantheon+ (1701 light curves, 1550 unique SNe, $0.001 < z < 2.26$).
4. *Tension dataset*: Adding SH0ES H_0 prior [3] + DES Y3 S_8 [5].

2. MCMC configuration

Parameter estimation uses Cobaya [22] (v3.5 original, v3.6.1 verification) with CAMB, convergence criterion $\hat{R}-1 < 0.01$, and > 1000 effective samples per parameter. The theory engine is unmodified stock CAMB; curvature is fixed to $\Omega_k = 0$. The MCMC analysis is phenomenologically equivalent to any ΛCDM+$\Delta N_{\rm eff}$ extension—the tension reduction is a property of the parameter space, not uniquely of spin-torsion cosmology.

3. Results

MCMC fits to the full tension dataset yield:

$$H_0 = 69.2 \pm 0.8 \text{ km s}^{-1} \text{ Mpc}^{-1}, \tag{5}$$

$$\sigma_8 = 0.785 \pm 0.016, \tag{6}$$

$$S_8 = 0.80 \pm 0.02. \tag{7}$$

These partially reduce the H_0 tension (from 4.9σ to 2.9σ vs. SH0ES) and the S_8 tension (from $\sim 3\sigma$ to $\sim 1\sigma$ vs. weak lensing).

4. Independent verification

Independent verification using Cobaya v3.6.1 with Planck NPIPE CamSpec TTTEEE + lowl TT/EE + lensing has produced two frozen dataset combinations with publication-quality convergence:

TABLE I. Independent verification results (Cobaya v3.6.1, CAMB v1.6.5). Posterior means $\pm 1\sigma$.

Parameter	**Full-tension**	**Planck+BAO+SN**
H_0 [km/s/Mpc]	67.68 ± 1.06	67.79 ± 1.09
$\Delta N_{\rm eff}$	-0.020 ± 0.169	$+0.065 \pm 0.17$
σ_8	0.803 ± 0.008	0.812 ± 0.009
S_8	0.814 ± 0.008	0.831 ± 0.018
Ω_m	0.308 ± 0.005	0.312 ± 0.006
Chains	6	6
Total samples	176,840	132,949
Worst $\hat{R} - 1$	0.001	0.003
Min ESS	4,744	4,692

Both datasets find $\Delta N_{\rm eff}$ consistent with zero. The H_0 values (67.68 and 67.79) are consistent with Planck ΛCDM (67.36 ± 0.54) at 0.3σ, confirming that the $\Delta N_{\rm eff}$ extension alone does not resolve the Hubble tension—the tension reduction in the original analysis (Eqs. 5–7) is driven by the SH0ES prior. The difference between the original central value ($H_0 = 69.2$, Eqs. 5–7) and the verification ($H_0 = 67.68$, Table I) reflects different Planck likelihood implementations (`plikHM` vs. `CamSpec`) and dataset vintages; both values lie within their respective 1σ uncertainties and agree at 1.1σ.

Comparison with independent EC torsion analyses.—Liu *et al.* [23] constrained an EC torsion model using DESI DR2 + PantheonPlus + DES Y5 + Planck, finding $H_0 = 68.41 \pm 0.32$ with torsion preferred over ΛCDM by AIC (ΔAIC $= -5.7$ to -6.6). Our verification agrees at 0.5σ in H_0 and 0.4σ in σ_8, providing independent cross-validation of the EC cosmological framework.

5. Bayesian model comparison

The Bayes factor is dataset-dependent: $\ln B = -1.2$ (Planck+BAO), $+2.1$ (Planck+BAO+SH0ES), $+4.8$ (full tension). The preference emerges specifically when tension data are included, because the extra parameter $\Delta N_{\rm eff}$ partially reduces these tensions. With Planck+BAO alone, the Occam penalty slightly disfavors the extension. This dataset dependence is expected and not a weakness: it reflects the model's statistical flexibility rather than unique predictive power.

TABLE II. Bayesian model comparison (effective $\chi^2_{\rm eff} \equiv -2\ln\mathcal{L}_{\rm max}$ for the full tension dataset).

Model	k	$\chi^2_{\rm eff}$	$\ln B$
ΛCDM	6	1156.2	0.0
wCDM	7	1154.8	−0.5
ECH phenom.	**7**	**1148.3**	**+4.8**

C. Observational consistency checks

The framework accommodates several observational features as consistency checks—not predictions.

1. Cosmic birefringence

The framework's parity-odd structure is qualitatively consistent with the 2.4–2.9σ cosmic birefringence detections from Planck [18] and ACT DR6 [19] (combined: $\beta = 0.242^\circ \pm 0.061^\circ$, 3.9σ). However, the minimal ECH framework has *no derived photon-torsion coupling*: matching the observed β requires an assumed coupling with $\mathcal{O}(1)$ coefficient, generic to any ALP model (see Route S1 closure, Sec. V D). This is a consistency check, not a prediction.

2. Galaxy spin asymmetry

A galaxy spin dipole asymmetry ($A_0 \sim 0.003$) is observed empirically, but the CMB isotropy bound constrains the framework's rotation-driven amplitude to $A_0 \lesssim 10^{-12}$—an order-of-magnitude gap that prevents any first-principles connection. The signal is empirical; its relation to the framework remains qualitative at best.

3. Fine-tuning reduction

A Monte Carlo sensitivity scan (10^5 samples) confirms that the scaling ansatz Eq. (4) reduces fine-tuning from 10^{120} to $\sim 10^5$: the viable range $N_{\rm tot} \in [79, 95]$ produces ρ_Λ within two orders of magnitude of the observed value, with $N_{\rm tot}$ as the single controlling parameter ($|\rho_s| = 0.996$). The residual tuning is concentrated in a single initial condition with potential dynamical answers through the bounce-to-inflation transition.

TABLE III. Fine-tuning comparison.

Model	Natural Scale	FT Score
ΛCDM	$M_{\rm Pl}^4$	$\sim 10^{120}$
Quintessence	$M_{\rm EW}^4$	$\sim 10^{60}$
$f(R)$ Gravity	H_0^4	$\sim 10^{40}$
ECH phenom.	$(\alpha/M)\mathcal{D}_{\rm inf}\, M_{\rm Pl}^4\,^\dagger$	$\sim \mathbf{10^5}$

†The ECH natural scale is a parametric estimate incorporating the scaling ansatz (4), not an independently derived scale; the fine-tuning score reflects sensitivity to $N_{\rm tot}$ within ∼4 e-folds.

4. Limit behavior

Internal consistency is verified: $T^{abc} \to 0$ recovers GR, $\mathcal{D}_{\rm inf} \to 1$ reproduces the standard CC problem, and $\gamma \to \infty$ recovers ECSK theory.

IV. THE DERIVATION PROGRAM

The central question is whether late-time $w = -1$ can be derived from first principles within the minimal ECH+Dirac model. We tested four routes, each with predefined gates and kill criteria frozen before computation [21].

A. Four routes tested

Track B (NJL condensate):
Test whether a pseudoscalar fermion condensate $\langle \bar\psi i\gamma^5\psi\rangle$ forms via the Nambu-Jona-Lasinio mechanism from the torsion-induced four-fermion interaction and persists as late-time vacuum energy.

Branch G v1 (strict one-loop effective action):
Compute the one-loop gravitational effective action $\Gamma[e]$ after torsion elimination and fermion integration, and test for γ-dependent vacuum content beyond standard cosmological-constant renormalization.

Route T1 (dynamical Immirzi field):
Promote $\gamma \to \theta(x)$ to a dynamical pseudoscalar and assess whether the reduced action contains novel gravitational content not reproducible by adding a standard axion-like particle (ALP) to GR.

Route S1 (parity-sensitive CMB phenomenology):
Assess whether the framework's parity-odd operator maps to distinctive, testable birefringence observables distinguishable from generic ALP predictions.

All four routes were closed with clean negative results. The detailed closure documentation is in the companion technical note [21]. Section V summarizes the results and Sec. VI extracts the structural lessons.

V. CLOSURE RESULTS

A. Track B: condensate route

Fierz rearrangement of the four-fermion interaction (3) decomposes it into standard NJL channels. The scalar/pseudoscalar coupling is

$$G_{\rm SP} = \frac{3\kappa^2}{16}\,\frac{\gamma^2-1}{\gamma^2+1}, \tag{8}$$

which is *repulsive* (negative) for $\gamma < 1$. At $\gamma = 0.274$, the NJL gap equation has no nontrivial solution. Even for $\gamma > 1$ (attractive sign), the coupling is $\sim 175\times$ below the NJL critical coupling, and gravitational catalysis is exponentially suppressed ($M_* \sim e^{-2100}\, M_{\rm Pl}$). Track B closes at Gate 1: no condensate forms.

B. Branch G v1: strict one-loop

After torsion elimination, the Dirac operator is $\not{D} = i\gamma^\mu \mathring{\nabla}_\mu$—the standard γ-independent Levi-Civita operator. The four-fermion term is quartic in ψ and does not contribute at one loop. The Seeley-DeWitt coefficients a_0, a_1, a_2 are the standard textbook values with no γ-dependence [15]. Branch G v1 closes: no novel Holst content at strict one-loop.

C. Route T1: dynamical Immirzi field

Promoting $\gamma \to \theta(x)$, torsion remains algebraic. After elimination, the reduced action is [11, 16]

$$S_{\rm red} = S_{\rm EH} + S_{\rm Dirac} + \int \big[-\tfrac{1}{2} f(\theta)(\partial\theta)^2 + \mathcal{L}_{\rm 4f} \big], \tag{9}$$

where $\mathcal{L}_{\rm 4f} = -(3\pi G/2)(\bar\psi\gamma^\mu\gamma^5\psi)^2$ is θ-*independent*. Every term is reproducible by adding a standard ALP to GR. The Nieh-Yan 4-form is exact (total derivative) in 4D; its topological nature ensures the gravitational origin washes out [16]. Cosmological dynamics give $w = +1$ (stiff matter) [17], the opposite of dark energy. No mechanism generates $V(\theta) \sim \Lambda$. Route T1 closes.

D. Route S1: parity CMB phenomenology

The minimal ECH action has no photon-polarization coupling. The parity-odd operator $\varepsilon^{abcd} K_{ab} R_{cd}$ is purely gravitational; the electromagnetic field strength does not appear in the Holst term or the reduced action. Any assumed coupling produces constant-β birefringence with two free parameters for one observable—identical to generic ALP phenomenology, with no distinctive spin-torsion signature. Route S1 closes.

TABLE IV. Four tested routes and outcomes.

Route	Target	Method	Failure
Track B	Condensate	Fierz + NJL	$G_{\rm SP} < 0$ at $\gamma < 1$
G v1	$\Gamma[e]$	1-loop	$\not{D}$ is γ-free
T1	$\theta(x)$	Lit. review	= standard ALP
S1	Birefr.	Assessment	No photon coupling

E. Summary

VI. STRUCTURAL LESSONS

The closures are not independent accidents. They reveal a hierarchy of structural obstructions that progressively narrows the space of viable geometric dark-energy models. Lessons 1–5 emerge from the minimal ECH model and establish that algebraic torsion cannot carry information into the IR. Lesson 6 emerges from the PGT extension and establishes that even propagating torsion can fail—through a different mechanism—if the mass and coupling share a common origin.

A. Algebraic torsion washes out

In the minimal ECH model, torsion is non-propagating. Its equation of motion is algebraic: the contorsion K is determined pointwise by the fermion spin density. Torsion elimination is exact and complete—no torsion-sector information survives into the reduced action except through the four-fermion coupling constants. Any mechanism that requires torsion to carry information into the infrared will fail in this model class.

B. Fixed γ leaves no IR fingerprint

γ enters the reduced action only through $G_{\rm eff}$ and $G_{\rm SP}$ in the four-fermion interaction. These are UV-scale contact couplings. At one loop, the fermion determinant is γ-independent. At the condensate level, the coupling is subcritical. γ sets the UV-scale spin-torsion structure but does not generate low-energy phenomena.

C. Dynamical γ reduces to generic ALP

The Nieh-Yan 4-form $N_4 = d(e^I \wedge T_I)$ is exact in 4D. Any coupling of a pseudoscalar θ to N_4 reduces to boundary terms plus standard kinetic/coupling structures after torsion elimination. The gravitational origin is topologically protected from leaving a local dynamical imprint. This is why Route T1 collapses to a generic ALP.

D. Parity alone is not a mechanism

The ECH action's parity-odd sector is real, but without a derived coupling to observable sectors (photons, matter), it produces no testable prediction distinguishing the framework from generic alternatives. Parity is a structural property of the theory, not by itself a mechanism or signal source.

E. No scale protection in the minimal model

None of the tested routes addresses naturalness. The condensate route produces exponentially suppressed scales but the wrong sign. The one-loop route has standard renormalization. The dynamical Immirzi route requires $V(\theta_0) = \Lambda_{\rm obs}$ by hand. Any successor model must confront scale naturalness from the start.

F. The mass-coupling lock

Even when torsion is promoted from algebraic to propagating (PGT), a new obstruction emerges: the parameter that controls the torsion mass also controls the torsion-matter coupling, and both go to zero together. In the ghost-free 0^- axial torsion model, $m_B \propto 1/\sqrt{|t_3|}$ and $g_{\rm eff} \propto 1/\sqrt{|t_3|}$, so making the mode cosmologically light automatically makes it cosmologically inert.

This is not a PGT-specific accident. The canonical-normalization relation that produces this locking is standard field theory; what is new is recognizing that it constitutes a binding constraint on geometric dark-energy mechanisms—specifically, that the ghost-free PGT torsion mode becomes cosmologically inert in exactly the parameter regime required for dark energy. The *mass-coupling lock* occurs generically when a single parameter controls both the kinetic normalization and the mass of a geometric mode. Consider a field ϕ with Lagrangian $\mathcal{L} = -\frac{1}{2}Z(\partial\phi)^2 - \frac{1}{2}\mu^2\phi^2 + g\,\phi\,J$. Canonical normalization $\phi_{\rm can} = \sqrt{Z}\,\phi$ gives $m = \mu/\sqrt{Z}$ and $g_{\rm eff} = g/\sqrt{Z}$: mass and coupling are locked. In PGT, $Z \propto |t_3|$ and $\mu^2 \propto M_{\rm Pl}^2$, producing exactly this structure. The same pattern appears in Brans-Dicke theory, where the scalar decouples as $\omega_{\rm BD} \to \infty$ (the GR limit *is* the decoupling limit), and in any geometric theory where the propagating mode inherits its kinetic term from the gravitational action without an independent mass-generating mechanism.

The lock is *evaded* when mass and coupling are controlled by independent parameters—for instance, through a Higgs-like mechanism (separate VEV sets the mass), a gauge symmetry (coupling is fixed by a charge), or an environmental mechanism (mass depends on background, not on kinetic normalization). In the Standard Model, the W boson mass $m_W = gv/2$ is set by *both* the gauge coupling g and the Higgs VEV v; taking $m_W \to 0$

by $v \to 0$ does not suppress g. No analogous structure exists in standard PGT.

Any next-generation geometric dark-energy model must avoid this lock: the mass and the coupling to observable sectors must be controlled by independent parameters, or the light-mass limit must not simultaneously decouple the field from matter.

VII. REQUIREMENTS FOR NEXT-GENERATION FOUNDATIONS

The six structural lessons translate into necessary conditions that any viable geometric dark-energy theory must satisfy simultaneously. These are not aspirational guidelines; each corresponds to a demonstrated failure mode. A candidate model that violates any single requirement is excluded by the same mechanism that closed the corresponding route.

The first four requirements (DR1–DR4) were formulated from the minimal-model closures; DR5 emerged from the Foundation A (PGT) assessment and addresses a structural obstruction not visible at the minimal level.

DR1: Survives reduction without washing out.: The model must contain at least one degree of freedom or structural feature that is present in both the full theory and the reduced/effective theory after all algebraic constraints are solved. *Test:* write down the reduced action. Does the mechanism's key ingredient still appear? *Closed by:* Lessons 1–3 (algebraic torsion, fixed γ, Nieh-Yan topology).

DR2: Addresses tiny-scale naturalness explicitly.: The model must contain an explicit statement of why the relevant scale is small: shift symmetry, discrete symmetry, gauge symmetry, radiative stability, dynamical relaxation, or sequestering. "The scale is a free parameter" is not acceptable. *Test:* if asked "why 10^{-122}?", does the model have a structural answer? *Closed by:* Lesson 5 (no scale protection in minimal model), Foundation A Phase 2B (no symmetry protects PGT torsion mass).

DR3: Offers unique mechanism or distinctive observable.: The model must produce at least one prediction not reproducible by ΛCDM+generic ALP. If observationally identical to ΛCDM+ALP, the model has no scientific value regardless of theoretical motivation. *Test:* name the distinctive prediction. Can a generic ALP reproduce it? *Closed by:* Lesson 4 (parity alone is not a mechanism), Foundation A Phase 2A (distinctive signatures suppressed by decoupling limit).

DR4: Fails cleanly if it doesn't work.: The model must be testable within a defined scope, with gates and kill criteria specified before computation. Negative results must be documentable. *Test:* can you write the gate structure before calculating?

DR5: Avoids the mass-coupling lock.: The model's IR-relevant degree of freedom must not decouple from observable sectors in the light-mass limit. Specifically, the mass and the coupling to matter (or to photons, gravitational waves, or other observable channels) must be controlled by *independent* parameters, or the light-mass limit must preserve the coupling at observable strength. A theory in which $m \to 0$ implies $g_{\rm eff} \to 0$ produces a cosmologically inert field regardless of its theoretical elegance. *Test:* compute $g_{\rm eff}(m)$ in the canonically normalized theory. Does $g_{\rm eff}$ remain at gravitational strength or above as $m \to H_0$? *Closed by:* Lesson 6 (mass-coupling lock in PGT).

Foundation A (PGT propagating torsion) passes DR1 and DR4 cleanly. It fails DR2, DR3, and DR5. These failures are not independent: the mass-coupling lock (DR5) is the structural mechanism by which DR3 fails at cosmological parameter values. This pattern—a theory that is healthy in the UV but observationally empty in the required IR limit—is generic to theories where the mass and coupling share a common kinetic-normalization origin (Sec. VI F). Evading DR5 requires an independent mass-generating mechanism, not merely a new choice of gauge group or field content.

VIII. CANDIDATE FOUNDATION DIRECTIONS

Three directions were identified that each address at least one structural lesson from Sec. VI. Foundation A (propagating torsion) has been tested through Phase 2 and found phenomenologically empty (Sec. VIII A). Foundations B and C remain untested starting points for future assessment using the methodology of Sec. IV; both must now additionally satisfy DR5 (mass-coupling lock avoidance).

A. Foundation A: Propagating torsion (tested)

The most natural upgrade to the minimal ECH model is Poincaré gauge theory (PGT), in which the spin connection is dynamical and torsion propagates. This directly eliminates Lesson 1 (algebraic torsion wash-out) by construction. We have conducted a systematic two-phase assessment of this direction.

Phase 1 result. The quadratic PGT action (10 free parameters: 3 torsion-squared, 6 curvature-squared, plus Λ_0) admits three ghost-free single-mode models, confirmed by independent analyses spanning four decades [28, 29]. The cleanest candidate is Model B: a 0^- pseudoscalar axial torsion mode, which is ghost-free, parity-odd, and couples to the fermion axial current. Its mass is $m_B = M_{\rm Pl}/(4\sqrt{\pi|t_3|})$, where t_3 is the dimensionless torsion-squared coupling. A cosmologically light mass ($m_B \sim$ meV) requires $|t_3| \sim 10^{58}$—the hierarchy is transferred to a dimensionless coupling, not solved.

Phase 2 result: the mass-coupling lock. The critical finding concerns what happens in the cosmologically interesting regime $|t_3| \gg 1$. The canonically normalized matter coupling is

$$g_{\text{eff}} \sim \frac{1}{M_{\text{Pl}}\sqrt{|t_3|}}\,, \tag{10}$$

so that m_B and g_{eff} decrease *together*: their ratio $m_B/g_{\text{eff}} \sim M_{\text{Pl}}^2$ is fixed. At $|t_3| \sim 10^{58}$ (meV-scale mass), the matter coupling is $\sim 10^{29}$ times weaker than gravitational strength. The theory is perturbatively consistent (no unitarity violation, no strong-coupling problem—the unitarity scale *rises* as $\sqrt{s_{\text{unit}}} \sim M_{\text{Pl}}\sqrt{|t_3|}$), but the torsion mode cannot thermalize, cannot produce detectable fifth forces, and cannot generate observable birefringence through its matter coupling.

Furthermore, no symmetry within PGT protects m_B: five candidate symmetries (gauge, shift, chiral, topological, supersymmetric) were investigated and excluded; the 't Hooft naturalness criterion fails ($m_B = 0$ does not restore a symmetry); and graviton loops generate $\delta m_B^2 \sim M_{\text{Pl}}^2/(16\pi^2)$, requiring fine-tuning of 1 in 10^{57} for $m_B \sim$ meV.

Structural lesson. Large $|t_3|$ is a *decoupling limit*, not a strong-coupling limit. The same parameter that makes the mass cosmologically light also kills the distinctive observational signatures. The reduced action is GR + massive Proca pseudovector (structurally richer than GR + ALP, with 3 DOF and axial-current coupling), but all distinctive features are suppressed to irrelevance at cosmological parameter values. The mode becomes indistinguishable from generic quintessence—only the Lagrangian knows it is torsion.

DR assessment (post-test): DR1 (survives reduction): **pass**—the 0^- mode propagates and is ghost-free. DR2 (scale naturalness): **fail**—no symmetry protection, graviton loop destabilizes mass. DR3 (distinctive observable): **fail**—matter coupling too suppressed at cosmological $|t_3|$. DR4 (clean failure): **pass**—the decoupling limit is a clean, physically interpretable negative result.

Verdict: Foundation A is structurally viable but phenomenologically empty for dark energy. It resolves the algebraic wash-out problem (Lesson 1) but encounters a new obstruction: the *mass-coupling lock* (Lesson 6, Sec. VI F).

B. Foundation B: Symmetry-protected geometric pseudoscalar

The dynamical Immirzi field failed because the Nieh-Yan density is exact in 4D. But in metric-affine gravity (where the connection is not metric-compatible), the Nieh-Yan form is *not* an exact differential. A pseudoscalar coupled to this non-topological Nieh-Yan density could retain geometric fingerprints after reduction, protected by a shift symmetry that keeps its mass small.

Why it avoids old failures: ALP collapse (Lesson 3) is avoided if the non-topological coupling retains geometric information. Scale protection (Lesson 5) is provided by the shift symmetry.

Biggest risk: This may be Route T1 in better clothing. The burden is entirely on demonstrating that the metric-affine Nieh-Yan coupling is genuinely non-topological and retains non-generic content after reduction.

First check: Is the Nieh-Yan density non-topological in metric-affine gravity? This is a precise mathematical question with a checkable answer.

DR assessment: DR1: depends on the first-check answer. If the Nieh-Yan coupling reduces to a total derivative after non-metricity is eliminated, this is Route T1 again. DR2: shift symmetry provides natural scale protection. DR3: if the non-topological coupling retains geometric content, the pseudoscalar is non-generic (different from standard ALP). DR4: yes—the topological/non-topological question is mathematically definite.

C. Foundation C: Vacuum sequestering

Instead of generating a vacuum energy of the right magnitude, one can attempt to cancel or sequester vacuum energy dynamically. The Kaloper-Padilla mechanism enforces, via global constraints in the action, that radiative corrections to the cosmological constant do not gravitate.

Why it avoids old failures: Scale naturalness (Lesson 5) is addressed directly. The small scale is protected by a structural constraint, not generated by a local mechanism.

Biggest risk: Weinberg's no-go theorem [1] constrains local adjustment mechanisms. Sequestering evades this by being global (non-local), but this non-locality is controversial. Pure sequestering may set Λ small but predict no distinctive observable (failing DR3).

First check: Write down the Kaloper-Padilla action [27] in first-order EC gravity. Does the torsion sector modify the sequestering constraint?

DR assessment: DR1: sequestering is a global mechanism, not reducible by algebraic constraints—survives by construction. DR2: directly addresses scale naturalness—this is its primary motivation. DR3: questionable—pure sequestering may set Λ small but predict no observable beyond ΛCDM. This is the critical weakness. DR4: yes—whether torsion modifies the constraint is checkable.

IX. DISCUSSION

A. Current status of the framework

The spin-torsion dark-energy framework occupies an honest intermediate position. It is *more* than generic ΛCDM: the scaling ansatz (4) provides a motivated parametric story for the dark-energy scale, reducing fine-tuning by 115 orders of magnitude. The parity-odd structure is real and could connect to observed birefringence if a photon coupling is established.

It is *less* than a first-principles derivation: $w = -1$ is assumed, the scaling ansatz is not a mechanism, the observational signatures are consistency checks rather than predictions, and both the minimal model class and its most natural extension (PGT propagating torsion) have been systematically tested and found insufficient for distinct structural reasons.

B. The closures as scientific contribution

Negative results in theoretical physics are undervalued but scientifically important. The five-closure program (four minimal routes plus the PGT Foundation A assessment):

- Establishes that both the minimal model and its most natural extension fail, for distinct structural reasons.
- Maps the failure mechanisms: algebraic torsion wash-out (minimal model) and the mass-coupling lock (PGT extension).
- Identifies five concrete requirements (DR1–DR5) for any successor, with DR5 (mass-coupling lock avoidance) being a previously unrecognized constraint.
- Demonstrates a reproducible methodology for disciplined model testing that can be applied to future candidates.

C. Comparison with other programs

The structural lessons identified here are not specific to our framework. Any geometric dark-energy program based on minimal ECH gravity faces the same algebraic-torsion wash-out. We position this work relative to five classes of approaches.

a. Phenomenological dark energy. The ΛCDM+ΔN_{eff} extension analyzed here is statistically equivalent to any additional relativistic species model (sterile neutrinos, early dark energy). The framework adds theoretical motivation for expecting $\Delta N_{\mathrm{eff}} > 0$ (bounce particle production) and a scaling argument for the Λ scale, but the MCMC fits themselves are standard. Quintessence and $w(z)$CDM models achieve comparable or better tension reduction with different parameter trade-offs; our advantage is the fine-tuning reduction from 10^{120} to 10^{5}.

b. EC and Holst torsion cosmology. Popławski [12] invokes torsion-induced four-fermion condensates to source dark energy; our Track B closure shows the S/P channel is repulsive at $\gamma < 1$, and the coupling is 175× subcritical. Liu *et al.* [23] find EC torsion preferred over ΛCDM by DESI DR2 (ΔAIC ≈ -6), but their torsion coupling is phenomenological—the same first-principles gap applies. Legner, Handley & Barker [25] proposed Torsion Condensation (TorC) within Poincaré gauge theory, showing that propagating torsion allows a larger inferred H_0—directly supporting Foundation A (Sec. VIII A). Fabbri [26] showed that propagating torsion as an axial-vector field naturally violates energy conditions, supporting the bounce mechanism.

c. Parity violation in gravity. Freidel *et al.* [9] and Mercuri [10, 11] established the Holst term's parity-violating consequences and showed the Barbero-Immirzi parameter is renormalized at one loop. Shapiro & Teixeira [13] computed one-loop divergences with external currents. These works establish the existence of a parity-odd coefficient but do not extract a finite vacuum energy or trace it through inflationary dilution. Our closure program (Branch G) shows the strict one-loop effective action is γ-independent after torsion elimination—a sharpening of the limitations implicit in these earlier calculations.

d. Metric-affine and propagating torsion. In metric-affine gravity (non-metricity + torsion), the Nieh-Yan density is *not* an exact differential—potentially avoiding the topological wash-out that closed Route T1. In Poincaré gauge theory (PGT), the spin connection is dynamical and torsion propagates with massive spin-2 and spin-0 modes. The ghost-free parameter subspace has been mapped by Sezgin & van Nieuwenhuizen [28], Yo-Nester, and Karananas [29]. Our Foundation A assessment (Sec. VIII A) confirms that ghost-free propagating modes exist but reveals a mass-coupling lock: the same parameter that produces a cosmologically light mass also suppresses the matter coupling to irrelevance (Lesson 6). These approaches resolve Lesson 1 but encounter a new structural obstruction.

e. Vacuum sequestering. The Kaloper-Padilla mechanism [27] enforces via global constraints that radiative corrections to Λ do not gravitate. This evades Weinberg's no-go theorem [1] through non-locality. Whether torsion modifies the sequestering constraint in EC gravity is unexplored (Foundation C first check). Pure sequestering may set Λ small but predicts no distinctive observable, failing our DR3 requirement.

D. What would change this assessment

The framework's status would be upgraded if:

1. A geometric theory is found in which the light-mass limit does *not* decouple the field from matter—i.e.,

a theory that evades the mass-coupling lock identified in Foundation A.

2. A derived photon-torsion coupling is established, producing a non-generic birefringence prediction.

3. Non-perturbative or higher-loop effects are shown to generate γ-dependent vacuum structure beyond the tested orders.

4. An external development (new symmetry, UV completion) identifies a protected light geometric degree of freedom with independent mass and coupling parameters.

None of these outcomes is assumed or expected. The Foundation A result shows that simply upgrading from algebraic to propagating torsion is not sufficient: the structural requirements are more demanding than previously appreciated.

X. CONCLUSION

We have presented a phenomenological dark-energy framework rooted in Einstein-Cartan-Holst gravity, reported the systematic closure of four minimal first-principles routes to deriving its central assumption ($w = -1$ at late times), and assessed the most natural next-generation extension—propagating torsion in Poincaré gauge theory—finding it structurally viable but phenomenologically empty for dark energy.

The framework provides a motivated scaling ansatz reducing fine-tuning from 10^{120} to $\sim 10^5$, cosmological fits partially reducing the H_0 and S_8 tensions, and a parity-odd structure qualitatively consistent with observed birefringence (without a derived photon coupling). These phenomenological results survive all closures intact.

The systematic model testing establishes:

- The minimal ECH model fails because algebraic torsion washes out, the Nieh-Yan density is topological, and no scale protection exists (Lessons 1–5).

- Propagating torsion in PGT resolves the algebraic wash-out but encounters a mass-coupling lock: the parameter that produces a cosmologically light mass simultaneously suppresses the matter coupling to irrelevance, and no symmetry protects the mass (Lesson 6).

- Six structural lessons and five decision rules (DR1–DR5) constrain the space of viable geometric dark-energy completions.

Any future geometric dark-energy model must not only survive reduction and address scale naturalness, but must avoid the mass-coupling lock: the field must remain cosmologically relevant in the light-mass limit. This is a nontrivial requirement that eliminates the most natural extension of the minimal framework.

The derivation of late-time $w = -1$ from geometry and the identification of a distinctive observational signature remain open problems. The contribution of this work is threefold: a well-tested phenomenological model, a systematic closure of the five most natural derivation strategies, and a concrete set of structural requirements—especially the mass-coupling lock—that any future geometric dark-energy proposal must address.

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Appendix A: Notation and Conventions

Natural units $c = \hbar = 1$ throughout, with $8\pi G = M_{\rm Pl}^{-2}$. Metric signature $(-,+,+,+)$. Reduced Planck mass $M_{\rm Pl} = (8\pi G)^{-1/2} = 2.435 \times 10^{18}$ GeV. $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ with $(\gamma^5)^2 = 1$. The vacuum energy density is $\rho_\Lambda = \Lambda_{\rm eff}\, M_{\rm Pl}^2/(8\pi)$; when we write $\rho_\Lambda \approx (2.3\ \text{meV})^4$ this refers to the energy density. Greek indices $\mu, \nu, \ldots$ denote spacetime coordinates; Latin indices $a, b, \ldots$ denote tangent-space (Lorentz) indices. The Levi-Civita symbol ε^{abcd} is totally antisymmetric with $\varepsilon^{0123} = +1$.

Key symbols:

- γ—Barbero-Immirzi parameter (0.274 ± 0.020)
- K_{ab}—contorsion tensor
- $T^a{}_{bc}$—torsion tensor
- α/M—parity-odd coefficient (mass dimension -1)
- $[(\alpha/M)\, M_{\rm Pl}] \sim 10^{-2}$—dimensionless one-loop suppression factor
- $\mathcal{D}_{\rm inf}$—inflationary dilution factor ($\sim e^{-3N_{\rm tot}}$)
- $\Xi \equiv [(\alpha/M)\, M_{\rm Pl}]\, \mathcal{D}_{\rm inf}$—inflationary suppression factor; $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$
- $G_{\rm eff} = (3\kappa^2/16)\,\gamma^2/(\gamma^2+1)$—effective four-fermion coupling
- $G_{\rm SP} \propto (\gamma^2-1)/(\gamma^2+1)$—scalar/pseudoscalar channel coupling
- $\Lambda_{\rm eff}$—effective cosmological constant
- $\rho_{\rm crit}$—bounce critical density ($\simeq 0.27\, \rho_{\rm Pl}$)

Abbreviations: LQG (Loop Quantum Gravity), EC (Einstein-Cartan), ECH (Einstein-Cartan-Holst), PGT (Poincaré Gauge Theory), ALP (axion-like particle), NJL (Nambu-Jona-Lasinio), MCMC (Markov Chain Monte Carlo), AIC/BIC (Akaike/Bayesian Information Criterion).

Appendix B: Parameter Summary

[a]Original values from the tension dataset (includes SH0ES H_0 prior). Independent verification (Table I) finds $H_0 = 67.68 \pm 1.06$, $\sigma_8 = 0.803 \pm 0.008$, $\Delta N_{\rm eff} = -0.020 \pm 0.169$ (full-tension) and $H_0 = 67.79 \pm 1.09$, $\sigma_8 = 0.812 \pm 0.009$, $\Delta N_{\rm eff} = +0.065 \pm 0.17$ (Planck+BAO+SN).

Key parameter correlations: $r(H_0, \sigma_8) = -0.45$; $r(\alpha/M, \mathcal{D}_{\rm inf}) = -0.89$ (only their product Ξ is well-constrained). The strong anticorrelation means the framework predicts $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$ as a single effective parameter, not two independent ones.

Appendix C: Claims Classification

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TABLE V. Complete parameter classification. Status categories: *Fixed* (external input), *Ansatz* (phenomenological scaling), *Assumed* (not derived), *Fit* (MCMC), *Derived* (from other parameters).

Parameter	**Description**	**Value**	**Status**	**Notes**
Framework parameters				
γ	Barbero-Immirzi	0.274 ± 0.020	Fixed	LQG area spectrum
α/M	Parity-odd coeff.	$\sim 10^{-33}$ eV^{-1}	Fit	One-loop motivated
$N_{\rm tot}$	Total e-folds	~ 92	Assumed	Fitted to ρ_Λ
$\mathcal{D}_{\rm inf}$	Inflation dilution	$\sim e^{-3N_{\rm tot}}$	Derived	From $N_{\rm tot}$
Ξ	Infl. suppression	$\sim 10^{-123}$	Ansatz	$= [(\alpha/M)M_{\rm Pl}]\mathcal{D}_{\rm inf}$
w	Equation of state	-1	Assumed	Not derived
Standard cosmological parameters (MCMC)				
H_0	Hubble constant	69.2 ± 0.8 [a]	Fit	km/s/Mpc
σ_8	Clustering ampl.	0.785 ± 0.016 [a]	Fit	
S_8	$\sigma_8(\Omega_m/0.3)^{0.5}$	0.80 ± 0.02	Derived	
Ω_m	Matter density	0.310 ± 0.008	Fit	
$\Omega_b h^2$	Baryon density	0.02237 ± 0.00015	Fit	
n_s	Spectral index	0.9649 ± 0.0042	Fit	
τ	Optical depth	0.054 ± 0.007	Fit	
Extended parameters				
$\Delta N_{\rm eff}$	Extra species	~ 0 [a]	Fit	Phenomenological
Ω_k	Curvature	0	Fixed	Mandated by $N_{\rm tot}$
$(\omega/H)_0$	Vorticity	$< 5 \times 10^{-11}$	Bounded	CMB isotropy

TABLE VI. Claims classification: what is derived, what is assumed, what is fit, and what has been retired by the closure program. This table supersedes the classification in [20].

Claim	**Status**	**Basis**	**Notes**
Derived (structural results)			
ECH action structure	Derived	Standard EC + Holst	Background
Torsion is algebraic (minimal ECH)	Derived	EC field equations	Structural
Perfect-square four-fermion	Derived	Torsion elimination	Exact result
$G_{\rm SP} < 0$ at $\gamma < 1$	Derived	Fierz rearrangement	Track B closure
One-loop $\not{D}$ is γ-free	Derived	Torsion elimination	G v1 closure
Dyn. Immirzi = ALP	Derived (lit.)	Calcagni-Mercuri 2009	T1 closure
No photon coupling in min. ECH	Derived	Action structure	S1 closure
Nieh-Yan is topological in 4D	Derived (lit.)	Differential geometry	T1 closure
Phenomenological (ansatz, fit, or assumption)			
$\rho_\Lambda = \Xi\, M_{\rm Pl}^4$	Ansatz	Scaling argument	Not a derivation
$w = -1$ at late times	Assumed	Observational input	Not derived
$\Delta N_{\rm eff}$ from bounce	Phenomenological	Qualitative motivation	Consistent with 0
H_0, σ_8, S_8 values	Fit	MCMC	Standard analysis
Fine-tuning $10^{120} \to 10^5$	Parametric obs.	Sensitivity scan	Valid
Birefringence consistency	Consistency check	Assumed coupling	Not a prediction
Galaxy spin $A_0 \sim 0.003$	Empirical	Hierarchical Bayes fit	Not derived
Retired (closed by derivation/assessment program)			
Birefringence as prediction	**Retired**	S1 closure	No photon coupling
γ drives late-time physics	**Retired**	B, G, T1 closures	UV-scale only
Condensate mechanism	**Retired**	Track B closure	$G_{\rm SP} < 0$
$\Delta N_{\rm eff}$ as distinctive signal	**Retired**	Verification	Generic to any $N_{\rm eff}$ ext.
Galaxy spin as prediction	**Retired**	Gap analysis	A_0 not derivable
PGT torsion as DE mechanism	**Retired**	Foundation A	Mass-coupling lock